

An exponential volume gap between B_p^n and $B_{p,\infty}^n$ for every finite p

Candidate solution packet for arXiv:1906.04997

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Status. Candidate full solution; likely valid, pending human review. The proof is deterministic and asymptotic. Numerical checks are included only as sanity checks and are not used in the proof.

1 Source problem

The source is A. Doležalová and J. Vybírál, *On the volume of unit balls of finite-dimensional Lorentz spaces*, arXiv:1906.04997, later published in the Journal of Approximation Theory [1].

For $x \in \mathbb{R}^n$, write

$$|x|_1^* \geq |x|_2^* \geq \cdots \geq |x|_n^*$$

for the nonincreasing rearrangement of the absolute coordinates. For $0 < p < \infty$, put

$$B_{p,\infty}^n = \left\{ x : \max_{1 \leq k \leq n} k^{1/p} |x|_k^* \leq 1 \right\}, \quad B_p^n = \left\{ x : \sum_{i=1}^n |x_i|^p \leq 1 \right\},$$

and

$$R_{p,n} := \frac{\text{vol}(B_{p,\infty}^n)}{\text{vol}(B_p^n)}.$$

The source proves exponential growth of $R_{p,n}$ for $0 < p \leq 2$ and asks whether it holds for all $0 < p < \infty$ [1, p. 14, Theorem 9 and the paragraph preceding it].

We show that the ratio $R_{p,n}$ defined in (23) grows exponentially for $0 < p \leq 2$. Naturally, we also conjecture that the same is true for all $0 < p < \infty$, but we leave this as an open problem.

Theorem 9. *For every $0 < p \leq 2$, there is a constant $C_p > 1$, such that*

$$R_{p,n} \gtrsim C_p^n$$

with the multiplicative constant independent on n .

Proof. We give the proof for even n 's, the proof for odd n 's is similar, only slightly more technical. Let $\mathcal{B}_p^n \subset \mathbb{R}^n$ be the set of vectors $x \in \mathbb{R}^n$, which satisfy

Figure 1: The open problem in the source paper, page 14.

Transcription. “Naturally, we also conjecture that the same is true for all $0 < p < \infty$, but we leave this as an open problem.” Here “the same” means that $R_{p,n} \gtrsim C_p^n$ for some $C_p > 1$, with the multiplicative constant independent of n .

2 Main result

Theorem 1. *For every $0 < p < \infty$, with $q = 1/p$,*

$$\liminf_{n \rightarrow \infty} R_{p,n}^{1/n} \geq \Lambda_p := \frac{(1+q)^{1+q}}{\Gamma(1+q)} > 1.$$

Consequently, for every $1 < C_p < \Lambda_p$ there is $A_p > 0$ such that

$$R_{p,n} \geq A_p C_p^n \quad (n \geq 1).$$

In particular, the open problem in [1] has an affirmative answer.

3 Proof intuition

Scale the positive part of $B_{p,\infty}^n$ by $n^{1/p}$. Membership then says that the empirical upper tail of the coordinates is bounded by the Pareto tail t^{-p} : no more than nt^{-p} coordinates may exceed $t \geq 1$.

We seek a one-dimensional probability density with large differential entropy and a tail below this Pareto obstacle. The useful density is constant from 0 to

$$a = (p+1)^{1/p}$$

and equals pt^{-p-1} after a . Its survival function is the tangent line to t^{-p} before the contact point and equals t^{-p} after it. A slight contraction makes the tail constraint strict.

A finite partition of the positive line converts this distribution into coordinate “type classes.” Stirling’s formula says that the logarithmic volume of a type class is its discrete entropy plus the logarithms of the bin widths; fine partitions recover differential entropy. Strict tail slack ensures that every point in the type class obeys all weak- ℓ_p rank inequalities. The resulting entropy exceeds the normalized ℓ_p -ball entropy by

$$(1+q) \log(1+q) - \log \Gamma(1+q) > 0,$$

which is the required exponential gap.

4 A type-volume lemma

For $q = 1/p$, define the rescaled positive weak ball

$$W_n := \{y \in [0, \infty)^n : y_k^* \leq (n/k)^q, \quad 1 \leq k \leq n\}.$$

Thus, up to coordinate hyperplanes of measure zero,

$$\text{vol}(B_{p,\infty}^n) = 2^n n^{-qn} \text{vol}(W_n). \tag{1}$$

For a density g on $[0, \infty)$, write

$$\overline{G}(t) = \int_t^\infty g(s) ds, \quad h(g) = - \int_0^\infty g(s) \log g(s) ds.$$

Lemma 1 (Tail constraint to volume). *Suppose g is a probability density on $[0, \infty)$, $\int |g \log g| < \infty$, and for some $\beta < 1$,*

$$\overline{G}(t) \leq \beta t^{-p} \quad (t \geq 1).$$

Then

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \text{vol}(W_n) \geq h(g).$$

Proof. We give the finite-bin argument in detail.

First truncate and renormalize. Given $\varepsilon > 0$, choose $M > 1$ so large that, for

$$g_M(t) = \frac{g(t) \mathbf{1}_{[0, M]}(t)}{\int_0^M g(s) ds},$$

we have

$$h(g_M) \geq h(g) - \varepsilon$$

and

$$\overline{G}_M(t) \leq \beta_M t^{-p} \quad (t \geq 1)$$

for some $\beta_M < 1$. This is possible because $\int |g \log g| < \infty$ and the omitted mass tends to zero; explicitly, one may take $\beta_M = \beta / \int_0^M g$, which is less than one for all sufficiently large M .

Choose a finite partition

$$0 = b_0 < b_1 < \cdots < b_m = M$$

which includes 1 as an endpoint and is so fine above 1 that

$$\beta_M b_{j-1}^{-p} < b_j^{-p} \quad \text{whenever } b_{j-1} \geq 1. \quad (2)$$

It may also be chosen fine enough that, for

$$I_j = (b_{j-1}, b_j], \quad \pi_j = \int_{I_j} g_M(t) dt,$$

the histogram entropy satisfies

$$H(\pi) + \sum_{j=1}^m \pi_j \log |I_j| \geq h(g_M) - \varepsilon, \quad (3)$$

where $H(\pi) = -\sum_j \pi_j \log \pi_j$. Indeed, the left side is the differential entropy of the density which is constant on each I_j and has mass π_j ; refinement makes it converge to $h(g_M)$.

For all sufficiently large n , choose nonnegative integers n_j with

$$\sum_j n_j = n, \quad \frac{n_j}{n} \longrightarrow \pi_j.$$

Because all the inequalities in (2) are strict, the rounding may be chosen so that, with $N_j = \sum_{\ell=j}^m n_\ell$,

$$\frac{N_j}{n} \leq b_j^{-p} \quad \text{for every } j \text{ with } b_j > 1. \quad (4)$$

To see this, the limiting cumulative proportion is

$$\sum_{\ell=j}^m \pi_\ell = \overline{G}_M(b_{j-1}) \leq \beta_M b_{j-1}^{-p} < b_j^{-p}.$$

Let E_n be the set of vectors having exactly n_j coordinates in I_j for every j . Then $E_n \subset W_n$. Indeed, a coordinate in I_j has decreasing rank at most N_j . If $b_j \leq 1$, then

$$y_i \leq b_j \leq 1 \leq (n/N_j)^q.$$

If $b_j > 1$, (4) gives

$$y_i \leq b_j \leq (n/N_j)^{1/p} = (n/N_j)^q.$$

Thus all rank inequalities defining W_n hold.

The volume of this type class is

$$\text{vol}(E_n) = \frac{n!}{\prod_{j=1}^m n_j!} \prod_{j=1}^m |I_j|^{n_j}.$$

Stirling's formula therefore gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \text{vol}(E_n) = H(\pi) + \sum_j \pi_j \log |I_j| \geq h(g) - 2\varepsilon.$$

Since $E_n \subset W_n$ and $\varepsilon > 0$ was arbitrary, the lemma follows. $\square$

5 The tangent Pareto density

Set

$$a = (p+1)^{1/p}, \quad c = \frac{p}{(p+1)^{1+1/p}},$$

and define

$$f(t) = \begin{cases} c, & 0 \leq t \leq a, \\ pt^{-p-1}, & t > a. \end{cases} \quad (5)$$

This is a probability density, since

$$ca = \frac{p}{p+1}, \quad \int_a^\infty pt^{-p-1} dt = a^{-p} = \frac{1}{p+1}.$$

It is also continuous at a , because

$$pa^{-p-1} = c.$$

Let $\bar{F}$ be its survival function. For $t \geq a$, $\bar{F}(t) = t^{-p}$. For $0 \leq t \leq a$,

$$\bar{F}(t) = 1 - ct.$$

The function $t \mapsto t^{-p}$ is convex, and $1 - ct$ is its tangent line at $t = a$. Hence

$$\bar{F}(t) \leq t^{-p} \quad (t \geq 1). \quad (6)$$

For $0 < r < 1$, let

$$g_r(t) = r^{-1}f(t/r).$$

Then, for $t \geq 1$, (6) gives the strict tail estimate

$$\bar{G}_r(t) = \bar{F}(t/r) \leq r^p t^{-p}. \quad (7)$$

Thus Lemma 1 applies with $\beta = r^p < 1$. Scaling a density changes differential entropy by the logarithm of the scale, so

$$h(g_r) = h(f) + \log r. \quad (8)$$

It remains to calculate $h(f)$. Since

$$\int_a^\infty pt^{-p-1} \log t \, dt = a^{-p} \left(\log a + \frac{1}{p} \right),$$

and $\log c = \log p - (p+1) \log a$, direct substitution in (5) yields

$$\begin{aligned} h(f) &= -\log p + (p+1) \log a + \frac{1}{p} \\ &= (1+q) \log(1+q) - q \log q + q, \quad q = \frac{1}{p}. \end{aligned} \quad (9)$$

6 Proof of Theorem 1

The classical volume formula is

$$\text{vol}(B_p^n) = \frac{2^n \Gamma(1+q)^n}{\Gamma(1+qn)}. \quad (10)$$

Combining Lemma 1, (1), (8), and (10), and using Stirling's formula, gives

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{1}{n} \log R_{p,n} &\geq h(g_r) + q \log q - q - \log \Gamma(1+q) \\ &= (1+q) \log(1+q) - \log \Gamma(1+q) + \log r. \end{aligned}$$

This is true for every $0 < r < 1$. Letting $r \uparrow 1$ gives

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log R_{p,n} \geq \Delta_q := (1+q) \log(1+q) - \log \Gamma(1+q).$$

We check that $\Delta_q > 0$. If $0 < q \leq 1$, log-convexity of Γ and $\Gamma(1) = \Gamma(2) = 1$ give $\Gamma(1+q) \leq 1 < (1+q)^{1+q}$. If $q \geq 1$, the standard inequality

$$\Gamma(1+q) \leq q^q$$

gives the same conclusion. For completeness, the latter follows by differentiating $q \log q - \log \Gamma(q+1)$: the bound $\psi(x) < \log x$ implies that its derivative is at least

$$1 + \log q - \log(q+1) = 1 - \log(1+1/q) > 0$$

for $q \geq 1$, and the function vanishes at $q = 1$. The digamma bound follows, for example, from its standard positive-integral representation [2, Chapter 6].

Therefore

$$\liminf_{n \rightarrow \infty} R_{p,n}^{1/n} \geq e^{\Delta_q} = \frac{(1+q)^{1+q}}{\Gamma(1+q)} > 1.$$

Every $C_p < e^{\Delta_q}$ is consequently a lower exponential base for all sufficiently large n . Decreasing the multiplicative constant to cover the finitely many remaining dimensions proves the second assertion. $\square$

7 Verification notes

- The proof uses no numerical premise. The script `code/verify_rates.py` evaluates the source paper’s exact inclusion–exclusion recurrence at high precision, checks the tangent-tail inequality on a grid, and compares finite-dimensional logarithmic ratios with the claimed asymptotic lower base.
- The decisive deterministic point is Lemma 1. Its type classes use finitely many bins, and the strict contraction $r < 1$ absorbs integer rounding. No probabilistic law-of-large-numbers assertion is needed.
- The proof gives a lower bound for the exponential rate. It does not claim that the displayed lower bound is the exact limit.

8 Novelty check and limitations

Before promotion, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1906.04997, the paper title, and the phrases “volume ratio,” “weak ℓ_p ,” and “Lorentz unit ball.” No prior run result was found.

A bounded external search used the arXiv id, the authors, the exact title, $R_{p,n}$, and combinations of “weak- ℓ_p volume exponential” and “Lorentz volume conjecture.” It found the source article and later work on probabilistic geometry of other Lorentz balls, but no paper resolving this exact all- p volume-ratio question. This is not an exhaustive bibliographic proof of novelty.

The human reviewer should focus on: (i) the passage from tail slack to the cumulative integer constraints in Lemma 1; (ii) the scaling (1); and (iii) the entropy identity (9).

References

- [1] A. Doležalová and J. Vybíral, *On the volume of unit balls of finite-dimensional Lorentz spaces*, J. Approx. Theory **255** (2020), 105407; arXiv:1906.04997.
- [2] M. Abramowitz and I. A. Stegun (eds.), *Handbook of Mathematical Functions*, National Bureau of Standards, 1964.